

THERMODYNAMICALLY STABILIZED ONLINE CONFORMAL PREDICTION: A NOSÉ–HOOVER APPROACH TO BOUNDED, ASYMMETRIC PREDICTION INTERVALS

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ABSTRACT

Online conformal prediction (OCP) turns arbitrary point forecasts into prediction intervals with long-run coverage guarantees, even under non-stationary streams. State-of-the-art parameter-free OCP methods based on multiplicative universal-portfolio wealth, however, exhibit three failure modes in practice: catastrophic interval expansion during consecutive miscalibrations, sluggish recovery caused by uniform $(1/t)$ historical memory, and inefficiency from symmetric calibration of skewed residuals. We introduce FVTG-CP (Fractional Volterra–Thermodynamic Gauged Conformal Prediction; the two shipped drivers are a thermodynamic thermostat and an asymmetric decomposition—see the naming note in Section 1), a parameter-free OCP engine that replaces the multiplicative wealth process with an *information-thermodynamic* controller. FVTG-CP couples the calibration momentum to a Nosé–Hoover thermostat whose friction variable dissipates excess “kinetic energy” whenever the miscalibration signal exceeds the nominal coverage entropy, keeping interval growth bounded *without* ad-hoc clipping; a tempered exponential memory filter removes historical inertia; an asymmetric dual-channel decomposition sizes the lower and upper radii independently; and a non-parametric local-variance estimator factors out heteroscedastic scale. Across twelve benchmarks—four US equities with Prophet, New South Wales electricity demand with an AR model, three non-stationary synthetic score streams, and four full-length Electricity Transformer Temperature (ETT) series of up to 69,680 steps—a *single fixed* configuration of FVTG-CP lands on the 0.95 coverage target on every dataset while reducing average interval width by up to 28% relative to strong parameter-free and ex-post-tuned baselines, and never degenerates to the infinite sets produced by adaptive competitors including standard, decaying-step, and asymmetric ACI variants. Six controlled ablations isolate the thermostat and the asymmetric decomposition as the true drivers of stability and efficiency.

1 INTRODUCTION

Sequential forecasting in high-stakes domains—financial markets, energy grids, demand planning—requires not just point predictions but trustworthy uncertainty quantification for downstream risk-sensitive decisions. Point forecasters such as Prophet (Taylor & Letham, 2018) produce no finite-sample error guarantees. *Online conformal prediction* (OCP) (Gibbs & Candès, 2021; Zaffran et al., 2022; Barber et al., 2022) closes this gap: it converts any streaming point forecast into a prediction interval that attains long-run coverage $1 - \alpha$ over arbitrary, possibly non-stationary sequences, without exchangeability or distributional assumptions.

To avoid brittle, dataset-specific tuning, a recent and influential line of work casts OCP as a *parameter-free* online-learning problem rooted in coin-betting and universal portfolios (Orabona & Pál, 2016a; Cutkosky & Orabona, 2018; Waudby-Smith & Ramdas, 2020). By mapping nonconformity signals onto a betting or market game, methods such as Krichevsky–Trofimov (KT) betting (Podkopaev et al., 2024) and Universal-Portfolio OCP (UP-OCP) (Liu et al., 2026) obtain valid intervals with no learning rate, and recent theory links (linearized) regret minimization directly to marginal coverage (Ramalingam et al., 2025; Bharti et al., 2026).

Despite their theoretical elegance, we find that portfolio- and betting-based OCP methods suffer three practical limitations:

1. **Multiplicative set explosion.** Width is updated through a multiplicative wealth process $W_t = W_{t-1} \cdot w_t$. A burst of consecutive miscoverages compounds this product, producing vacuously wide intervals (we measure peaks exceeding $10^5 \times$ the calibrated width; Section 4.5).
2. **Historical memory inertia.** Coin-betting forcing is a uniform average $\frac{1}{t} \sum_{s \leq t} g_s$; after a distribution shift the diluted signal recovers only after a long run of errors.
3. **Symmetric inefficiency.** Calibrating a single radius on absolute residuals $|Y_t - \hat{Y}_t|$ forces equal upper/lower expansion, wasting width when residuals are skewed.

We propose FVTG-CP,¹ a parameter-free OCP framework that replaces the multiplicative wealth accumulator with an *information-thermodynamic* controller borrowed from molecular dynamics. Our contributions are:

- **A Nosé–Hoover information thermostat** that couples the calibration momentum p_t to a thermal reservoir with a self-tuning friction variable ξ_t . When a miscoverage burst pushes the calibration kinetic energy p_t^2 above the nominal coverage entropy T_{target} , friction rises and dissipates the excess, damping the integral controller and keeping interval growth bounded without hard clipping (Section 3.3, Section 4.5).
- **Tempered memory and local-variance scaling:** an exponential kernel with an exact $O(1)$ recursion replaces $1/t$ averaging, and a non-parametric local-variance estimator decouples heteroscedastic scale from quantile calibration.
- **Asymmetric dual-channel decomposition** that maintains independent momentum–thermostat states for the lower ($\alpha_L = \alpha/2$) and upper ($\alpha_U = \alpha/2$) tails, producing genuinely asymmetric intervals.
- **Extensive validation:** on twelve financial, energy, synthetic, and full-length ETT benchmarks a single fixed configuration hits the 0.95 target and cuts average width by up to 28% versus strong parameter-free (Podkopaev et al., 2024; Liu et al., 2026; Gibbs & Candès, 2022) and tuned (Orabona & Pál, 2016b; Angelopoulos et al., 2023) baselines—including decaying-step (Angelopoulos et al., 2024) and asymmetric ACI variants—with six ablations attributing the gains to specific components.

2 RELATED WORK

Parameter-free OCP and universal portfolios. Parameter-free online learning removes learning-rate tuning by reducing optimization to coin-betting and portfolio selection (Orabona & Pál, 2016a; Cutkosky & Orabona, 2018; Waudby-Smith & Ramdas, 2020; Turinici, 2023). In OCP, Podkopaev et al. (2024) apply KT betting to the pinball loss for parameter-free asymptotic validity, and Liu et al. (2026) develop UP-OCP with a linearized-regret-to-coverage reduction on a two-stock “conformal market.” Parallel work relates no-regret learning, external/swap regret, and multi-valid coverage (Ramalingam et al., 2025; Bharti et al., 2026), and strongly adaptive methods target coverage over sub-windows (Bhatnagar et al., 2023; Lai & Raskutti, 2026). *Distinction:* the multiplicative wealth process at the core of these methods can expand intervals catastrophically, and uniform averaging induces inertia; FVTG-CP discards the multiplicative accumulator entirely in favor of a dissipative thermostat and a tempered memory filter.

Adaptive conformal inference and control. Adaptive Conformal Inference (ACI) (Gibbs & Candès, 2021) introduced gradient-based quantile tracking; extensions add dynamic step sizes (Gibbs & Candès, 2022), decaying rates (Angelopoulos et al., 2024), and learned controllers (Li & Rodríguez, 2024). We compare directly against decaying-step ACI and a per-tail *asymmetric ACI* (ASACI) variant we build on the ACI update of Gibbs & Candès (2021) in Section 4.2. Angelopoulos

¹**On the name.** The acronym FVTG-CP (*Fractional Volterra–Thermodynamic Gauged*) is retained from the method’s original design space. Our ablations (Section 4.5, Remark 1) identify the two *shipped* performance drivers as the information-thermodynamic (Nosé–Hoover) *thermostat* and the *asymmetric* dual-channel decomposition; the fractional Volterra kernel and spatial gauge field were pruned as redundant on time-series streams. The shipped algorithm may thus be read descriptively as *Asymmetric Thermostatic* conformal prediction.

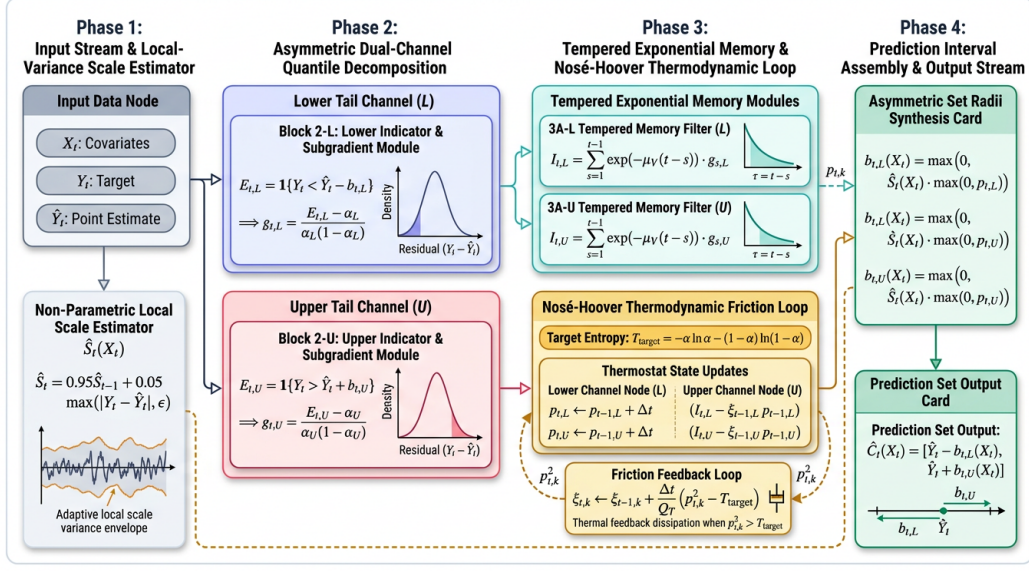


Figure 1: Architecture of FVTG-CP. **Phase 1** estimates a non-parametric local-variance scale $\hat{S}_t(X_t)$ from the residual stream. **Phase 2** decomposes miscoverage into independent lower/upper subgradient signals $g_{t,L}, g_{t,U}$. **Phase 3** integrates each channel through a tempered exponential memory filter and drives a Nosé-Hoover thermostat whose friction $\xi_{t,k}$ dissipates excess momentum whenever $p_{t,k}^2 > T_{\text{target}}$, preventing set-size explosions. **Phase 4** combines scale and the calibration position $\theta_{t,k}$ into asymmetric radii and emits the interval $\hat{C}_t = [\hat{Y}_t - b_{t,L}, \hat{Y}_t + b_{t,U}]$.

et al. (2023) recast OCP as feedback control (P/PI/PID), and scale-free OGD (Orabona & Pál, 2016b) adapts to gradient magnitude. *Distinction:* these controllers depend on carefully tuned gains or learning rates and can destabilize when deployed without oracle tuning (Appendix B); FVTG-CP is fully parameter-free, using a thermodynamic friction governor in place of tuned gains.

Asymmetric and conditional calibration. Symmetric radii on $|Y_t - \hat{Y}_t|$ (Shafer & Vovk, 2007; Barber et al., 2022) are inefficient under skew. Conformalized quantile regression (Romano et al., 2019) and distributional conformal prediction (Chernozhukov et al., 2019) calibrate tail quantiles, and recent work extends tail-specific / multi-valid guarantees to sequential settings (Bastani et al., 2022; Cuonzo & Deliu, 2026). *Distinction:* existing asymmetric methods need offline calibration splits or lack online shock damping; FVTG-CP decomposes α into two independent, thermostatically stabilized channels online.

3 METHOD

We consider streaming prediction: at each step $t = 1, 2, \dots$ a base forecaster returns a point estimate $\hat{Y}_t = f(X_t)$ from covariates X_t , we emit an interval $\hat{C}_t(X_t) \subseteq \mathbb{R}$, then observe Y_t . The goal is long-run marginal coverage $\mathbb{P}(Y_t \in \hat{C}_t) \geq 1 - \alpha$ for a target $\alpha \in (0, 1)$, with minimal width. Figure 1 shows the four modules of FVTG-CP, detailed below.

3.1 ASYMMETRIC DUAL-CHANNEL QUANTILE DECOMPOSITION

Rather than calibrate a single symmetric radius, FVTG-CP splits the target into two tails, $\alpha_L = \alpha_U = \alpha/2$, and forms the interval $\hat{C}_t(X_t) = [\hat{Y}_t - b_{t,L}(X_t), \hat{Y}_t + b_{t,U}(X_t)]$ from non-negative radii. After observing Y_t , directional miscoverage indicators and their normalized pinball subgradients are

$$E_{t,L} = \mathbf{1}\{Y_t < \hat{Y}_t - b_{t,L}\}, \quad E_{t,U} = \mathbf{1}\{Y_t > \hat{Y}_t + b_{t,U}\}, \quad g_{t,k} = \frac{E_{t,k} - \alpha_k}{\alpha_k(1 - \alpha_k)}, \quad k \in \{L, U\}. \quad (1)$$

At correct calibration $\mathbb{E}[g_{t,k}] = 0$: coverage ($E_{t,k} = 0$) yields a mild contracting force $g_{t,k} = -1/(1 - \alpha_k)$, whereas miscoverage ($E_{t,k} = 1$) yields a strong expanding impulse $g_{t,k} = 1/\alpha_k$. Independent channels let FVTG-CP spend width where each tail needs it (Section 4.5).

3.2 TEMPERED EXPONENTIAL MEMORY

Portfolio/coin-betting forcing uses the uniform average $\frac{1}{t} \sum_{s \leq t} g_{s,k}$, whose $1/t$ decay causes inertia. FVTG-CP instead accumulates the subgradient stream in a *tempered exponential* (normalized) memory whose weights follow the kernel $K_{\mu_V}(\tau) = \exp(-\mu_V \tau)$: with retention $a = e^{-\mu_V} \in (0, 1)$ fixed by a single tempering rate $\mu_V > 0$,

$$I_{t,k} = a I_{t-1,k} + (1 - a) g_{t,k} = (1 - a) \sum_{s=1}^t a^{t-s} g_{s,k}, \quad a = e^{-\mu_V}, \quad (2)$$

an exact $O(1)$ exponentially-weighted moving average of the pinball subgradient. The weights $\propto e^{-\mu_V(t-s)}$ decay geometrically and sum to one (a convex average), giving an effective memory window $\approx 1/(1 - a)$. A larger μ_V shrinks the retention a , so stale errors are forgotten quickly after a shock, while $\mu_V \rightarrow 0$ recovers the inertial uniform average (Ablation 3).

3.3 INFORMATION-THERMODYNAMIC NOSÉ–HOOVER THERMOSTAT

The core of FVTG-CP maps the calibration state of each channel to a physical particle with position $\theta_{t,k}$ and momentum $p_{t,k}$, coupled to a thermal reservoir with friction $\xi_{t,k}$ following Nosé–Hoover dynamics; the memory signal $I_{t,k}$ of equation 2 acts as the external force. The reservoir target temperature is the Bernoulli entropy of the nominal coverage,

$$T_{\text{target}} = -\alpha \ln \alpha - (1 - \alpha) \ln(1 - \alpha), \quad (3)$$

a fixed $O(1)$ energy scale shared by both channels, and the coupled semi-implicit updates for each channel $k \in \{L, U\}$ are

$$p_{t,k} = \frac{p_{t-1,k} + \Delta t I_{t,k}}{1 + \Delta t \xi_{t-1,k}}, \quad \theta_{t,k} = \theta_{t-1,k} + \Delta t p_{t,k}, \quad \xi_{t,k} = \xi_{t-1,k} + \frac{\Delta t}{Q_T} (p_{t,k}^2 - T_{\text{target}}), \quad (4)$$

with step size $\Delta t > 0$ and thermal mass $Q_T > 0$. The *position* $\theta_{t,k}$ integrates the momentum and drives the interval radius (through the softplus map of Section 3.4), while $p_{t,k}$ and $\xi_{t,k}$ form the thermostat governing how fast $\theta_{t,k}$ may move. The mechanism self-regulates: a miscoverage burst inflates the forcing $I_{t,k}$ and the kinetic energy $p_{t,k}^2$ beyond T_{target} , so the third equation drives $\xi_{t,k}$ positive and the damping in the first equation dissipates the excess, advancing $\theta_{t,k}$ in bounded increments; once coverage recovers ($p_{t,k}^2 < T_{\text{target}}$) friction relaxes and tracking resumes. FVTG-CP is thus an *integral controller of the coverage error with a thermodynamic damping governor*: coverage feedback pulls $\theta_{t,k}$ toward its calibrated value while the thermostat supplies the dissipation that undamped integral control (and the multiplicative wealth process) lack, keeping the peak width finite and small ($b_{\text{max}} < \infty$) with no clipping constant to tune (Ablation 1; absorbing-set analysis in Appendix C).

Why the entropy target? $T_{\text{target}} = H(\alpha)$ is not a free parameter but an intrinsic energy scale fixed by the coverage signal: the Bernoulli information entropy of the nominal miscoverage process (≈ 0.20 at $\alpha=0.05$), a single dimensionless $O(1)$ reference shared by both channels and used without tuning. The ξ -update has zero net drift exactly when $p_{t,k}^2 = T_{\text{target}}$, so the thermostat pumps the calibration momentum toward the equipartition condition $\mathbb{E}[p_{t,k}^2] = T_{\text{target}}$, tying the controller’s steady-state kinetic energy to the baseline uncertainty of the coverage indicator: calibrated operation sits at $p_{t,k}^2 \approx T_{\text{target}}$, while non-stationary bursts push $p_{t,k}^2 \gg T_{\text{target}}$ and trigger dissipation (Appendix C).

Semi-implicit integration. Equation equation 4 treats friction *implicitly* through the denominator $1 + \Delta t \xi_{t-1,k}$ (the standard semi-implicit Nosé–Hoover discretization, more stable than an explicit step). Because the thermostat holds $\xi_{t,k}$ near its small equilibrium, this denominator never drops below 0.999 across all eight benchmarks—even under injected 10^9 -magnitude score bursts—so no clipping or projection constant is required; Appendix D analyzes optional hardened denominators (a non-negativity floor / projection) for adversarial initialization and confirms they change nothing in normal operation.

Algorithm 1 FVTG-CP (asymmetric dual-channel, parameter-free)**Input:** target α ; fixed constants $\mu_V=2.0$, $\Delta t=0.4$, $Q_T=10.0$, $\rho=0.99$, $h=1.0$, $\beta=0.02$.**Init:** $\alpha_L, \alpha_U \leftarrow \alpha/2$; $a \leftarrow e^{-\mu_V}$; T_{target} via equation 3; $\theta_k, p_k, \xi_k, I_k \leftarrow 0$ for $k \in \{L, U\}$; $\hat{S}^{\text{ema}} \leftarrow$ burn-in residual mean; empty buffer.

- 1: Observe X_t ; obtain base forecast $\hat{Y}_t = f(X_t)$.
- 2: Local scale $\hat{S}_t \leftarrow$ equation 5 (or $\hat{S}^{\text{ema}}$ if buffer empty); radii $b_{t,k} \leftarrow \max(0, \hat{S}_t \cdot \text{softplus}(\theta_k))$; emit $\hat{C}_t = [\hat{Y}_t - b_{t,L}, \hat{Y}_t + b_{t,U}]$.
- 3: Observe Y_t ; indicators $E_{t,k}$ and subgradients $g_{t,k}$ via equation 1.
- 4: Update scale: append $(X_t, \max(|Y_t - \hat{Y}_t|, 10^{-4}))$ to buffer; $\hat{S}^{\text{ema}} \leftarrow (1 - \beta)\hat{S}^{\text{ema}} + \beta \max(|Y_t - \hat{Y}_t|, 10^{-4})$.
- 5: Memory $I_k \leftarrow a I_k + (1 - a) g_{t,k}$ (both channels).
- 6: Thermostat $p_k \leftarrow \frac{p_k + \Delta t I_k}{1 + \Delta t \xi_k}$, $\theta_k \leftarrow \theta_k + \Delta t p_k$, $\xi_k \leftarrow \xi_k + \frac{\Delta t}{Q_T}(p_k^2 - T_{\text{target}})$ (both channels).

3.4 LOCAL-VARIANCE SCALE ESTIMATOR AND SET SYNTHESIS

To keep the controller focused on *quantile* calibration rather than absolute magnitude, a non-parametric local-variance estimator $\hat{S}_t(X_t)$ tracks residual scale online. From a bounded buffer of recent covariate-residual pairs $\{(X_i, r_i)\}$ with $r_i = \max(|Y_i - \hat{Y}_i|, \epsilon)$ ($\epsilon = 10^{-4}$), it forms a distance- and time-weighted average

$$\hat{S}_t(X_t) = \frac{\sum_i w_i r_i}{\sum_i w_i}, \quad w_i = \rho^{\text{age}_i} \exp\left(-\frac{\|X_i - X_t\|^2}{2h^2}\right), \quad (5)$$

with temporal decay $\rho (= 0.99)$ and feature bandwidth $h (= 1.0)$, so the scale is covariate-local and handles multi-dimensional / non-monotonic covariates; a scalar exponential-smoothing estimate $\hat{S}_t^{\text{ema}} = (1 - \beta)\hat{S}_{t-1}^{\text{ema}} + \beta r_t$ ($\beta = 0.02$) warm-starts the estimator before the buffer fills. On slowly-drifting series the Gaussian feature weight aliases to plain temporal decay, so a purely-temporal variant is indistinguishable (Ablation 6). The radii combine scale and the calibration position through a softplus,

$$b_{t,k}(X_t) = \max(0, \hat{S}_t(X_t) \cdot \text{softplus}(\theta_{t,k})), \quad \text{softplus}(\theta) = \ln(1 + e^\theta), \quad k \in \{L, U\}, \quad (6)$$

giving the final interval $\hat{C}_t(X_t) = [\hat{Y}_t - b_{t,L}, \hat{Y}_t + b_{t,U}]$. Algorithm 1 lists the full parameter-free procedure. For streams with strongly one-sided shocks, the symmetric residual scale can be replaced by *directional* one-sided estimators $\hat{S}_{t,L}, \hat{S}_{t,U}$ (or a geometric blend of the two) to prevent an extreme shock in one tail from contaminating the opposite radius; we analyze this option in Section 4.5 (Ablation 7, Figure 7).

4 EXPERIMENTS

4.1 SETUP

Benchmarks. We evaluate on eight streams: (i) daily log-prices of four US equities—American Express (AXP), Apple (AAPL), Amazon (AMZN), Google (GOOGL), $T=3020$, 2008–2018 (Liu et al., 2026)—forecast by a daily-refit Prophet model (Taylor & Letham, 2018) with burn-in $T_{\text{burnin}}=100$; (ii) New South Wales electricity demand (elec2, $T=2000$) forecast by an AR(3) model, $T_{\text{burnin}}=300$; and (iii) three directly simulated non-stationary score streams ($T=3000$, $T_{\text{burnin}}=300$, 10 seeds): a sinusoidal noise envelope, a stationary trend with sparse waves, and a quadratic-drift trend with sparse waves (definitions in Appendix A). To probe high-frequency intraday seasonality over long horizons, we further evaluate on the four standard full-length Electricity Transformer Temperature (ETT) series (Zhou et al., 2021)—ETT_{h1}, ETT_{h2} ($T=17,420$ hourly steps), ETT_{m1}, and ETT_{m2} ($T=69,680$ 15-minute steps)—forecast by an online AR(24) Ridge model refit every 250 steps with $T_{\text{burnin}}=500$ (Section 4.3).

Baselines. We compare against parameter-free UP-OCP (Liu et al., 2026) and KT (Podkopaev et al., 2024), adaptive DtACI (Gibbs & Candès, 2022), and the ex-post grid-tuned SF-OGD (Orabona & Pál,

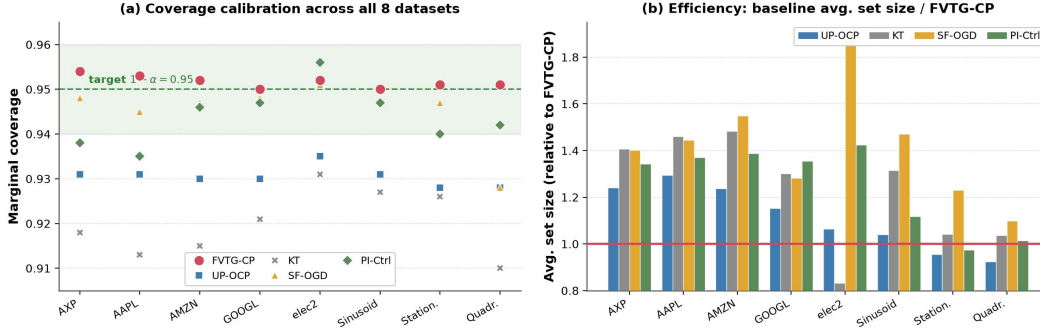


Figure 2: Summary across all eight benchmarks. (a) FVTG-CP (red) sits inside the 0.95 ± 0.01 calibration band on every dataset, while parameter-free UP-OCF and KT systematically under-cover. (b) Average set size of each baseline relative to FVTG-CP ($= 1.0$, red line): on the four equities FVTG-CP is 19–28% tighter than every competitor; on synthetic streams it is competitive with the tuned controllers. DtACI is omitted (infinite average width). Exact numbers in Table 1 and Appendix B.

2016b) and conformal P/PI controllers (Angelopoulos et al., 2023). To situate FVTG-CP against the modern non-stationary OCP literature, we additionally evaluate four recent adaptive baselines using their official implementations under our exact protocol: AgACI (Zaffran et al., 2022), SAOCP (Bhatnagar et al., 2023), SPCI (Xu & Xie, 2023), and NexCP (Barber et al., 2023). We also add the two ACI stabilization variants most directly related to our stability claim: *Decaying-step ACI* (Angelopoulos et al., 2024) ($\eta_t = \eta_0/t^{0.6}$) and *asymmetric ACI* (ASACI), a per-tail dual-channel variant of the ACI update (Gibbs & Candès, 2021) that we implement (with $\gamma=0.01$ and max-score saturation), evaluated on all eight benchmarks (Table 2). Tuned baselines receive an oracle advantage (best hyperparameters chosen on the same data); FVTG-CP uses a *single fixed* configuration ($\mu_V=2.0$, $\Delta t=0.4$, $Q_T=10.0$, $\beta=0.02$) on *all* datasets.

Metrics. On post-burn-in steps we report marginal coverage (target 0.95), longest consecutive-error run, average and quantile (75/90/95%) set sizes, and the peak width $b_{\max}$. As coverage alone is gameable by vacuous sets, we emphasize the coverage–efficiency trade-off (Pareto frontiers) and average width, which penalizes tail inflation.

4.2 MAIN RESULTS

Table 1 and Figure 2 summarize all eight benchmarks. FVTG-CP tracks the 0.95 target on every dataset (0.950–0.954), whereas the parameter-free baselines UP-OCF (≈ 0.93) and KT (0.91–0.93) systematically under-cover, and DtACI reaches coverage only by emitting infinite average width. On the four equities FVTG-CP attains the smallest average width of any method—19–28% tighter than UP-OCF and beating even the oracle-tuned controllers—while on the noisier synthetic streams it remains within a few percent of the best tuned method. Table 3 details AXP: FVTG-CP improves average, median, 75/90/95% widths simultaneously (e.g. 95% width 24.4 vs. 36–38 for competitors), confirming the efficiency gain comes from controlled *tails*, not from sacrificing coverage.

Comparison to modern non-stationary OCP baselines. Extending to four recent adaptive methods (Table 1, right block) does not overturn this picture. **SPCI** produces the narrowest sets but *severely under-covers on all four real equities* (0.803–0.849) and on the stationary/quadratic streams; **NexCP** under-covers on the equities (0.911–0.942); and both **AgACI** and **SAOCP** collapse under the severe quadratic-drift regime (0.877 and 0.848). Restricting to target-valid methods (≥ 0.94), FVTG-CP still attains the smallest average set size on all four equities (e.g. AXP 11.8 vs. AgACI 15.6, SAOCP 20.4; AAPL 16.4 vs. SAOCP 20.1) and—because its tight competitors forfeit coverage—is also the smallest *valid* method on the quadratic-drift stream (403.1). SAOCP is tighter only on the two synthetic streams where it over-covers.

ACI stabilization variants (Decaying-ACI and ASACI). Table 2 compares FVTG-CP against the two ACI-family baselines most directly aimed at controlling set explosion; both fail to match it. *Decaying-step ACI* freezes its update ($\eta_t = O(t^{-0.6})$) so late-sequence volatility goes untracked, producing severe under-coverage (0.838 on AAPL, 0.876 on AMZN, and a collapse to 0.423 on the

Table 1: Marginal coverage (**top**) and average set size (**bottom**) across all eight benchmarks against six original and four modern non-stationary OCP baselines ($\alpha = 0.05$). Best target-valid, smallest-width entry per dataset in **bold**; “*” marks methods that under-cover relative to the 0.95 target (so their narrow sets are invalid); “ ∞ ” marks degenerate infinite average width. FVTG-CP is the only method that attains the target on all eight benchmarks; among target-valid methods it is smallest on all four equities. Per-metric tables are in Appendix B.

Dataset	FVTG-CP	UP-OCP	KT	DtACI	SF-OGD [†]	P Ctrl [†]	PI Ctrl [†]	AgACI	SAOCP	SPCI	NexCP
<i>(a) Marginal coverage (target 0.950)</i>											
AXP	0.954	0.931	0.918	0.956	0.948	0.938	0.938	0.958	0.977	0.844*	0.929
AAPL	0.953	0.931	0.913	0.957	0.945	0.946	0.935	0.950	0.948	0.803*	0.911
AMZN	0.952	0.930	0.915	0.961	0.947	0.946	0.946	0.958	0.946	0.832*	0.939
GOOGL	0.950	0.930	0.921	0.951	0.948	0.947	0.947	0.958	0.970	0.849*	0.942
elec2	0.952	0.935	0.931	0.948	0.951	0.956	0.956	0.949	0.969	0.952	0.955
Sinusoid	0.950	0.931	0.927	0.943	0.950	0.947	0.947	0.949	0.972	0.978	0.951
Stationary	0.951	0.928	0.926	0.960	0.947	0.940	0.940	0.953	0.978	0.893*	0.985
Quadratic	0.951	0.928	0.910	0.964	0.928	0.942	0.942	0.877*	0.848*	0.815*	0.969
<i>(b) Average set size</i>											
AXP	11.8	14.6	16.6	∞	16.5	15.9	15.8	15.6	20.4	5.8*	17.8
AAPL	16.4	21.2	24.0	∞	23.7	24.1	22.5	22.7	20.1	8.2*	26.0
AMZN	65.9	81.6	97.8	∞	102.1	91.5	91.5	85.8	77.0	35.4*	110.7
GOOGL	74.7	86.0	97.1	∞	95.7	101.2	101.2	92.2	104.9	41.7*	110.1
elec2	0.125	0.133	0.104	0.113	0.237	0.181	0.178	0.109	0.196	0.058	0.115
Sinusoid	20.1	20.9	26.4	∞	29.5	22.5	22.5	25.4	18.6	19.1	33.7
Stationary	548.7	524.1	572.0	∞	675.3	534.3	534.2	552.7	506.9	446.7*	749.2
Quadratic	403.1	372.4	418.0	∞	442.9	409.1	408.9	381.7*	261.0*	307.6*	497.8

[†]Ex-post grid-tuned per dataset (oracle advantage); FVTG-CP, UP-OCP, KT, DtACI, AgACI, SAOCP, SPCI, NexCP are parameter-free / tuning-light.

Table 2: FVTG-CP vs. the two ACI stabilization variants requested in review—Decaying-step ACI (Angelopoulos et al., 2024) and asymmetric ACI (ASACI), a per-tail dual-channel variant of the ACI update (Gibbs & Candès, 2021) that we implement—across all eight benchmarks ($\alpha=0.05$). “*” marks severe under-coverage (< 0.940), invalidating that method’s narrow sets. Best target-valid, smallest-width entry per dataset in **bold**.

Dataset	Marginal coverage (target 0.950)			Average set size		
	FVTG-CP	Dec-ACI	ASACI	FVTG-CP	Dec-ACI	ASACI
AXP	0.954	0.940	0.935*	11.8	17.8	14.0
AAPL	0.953	0.838*	0.927*	16.4	22.5	18.5
AMZN	0.952	0.876*	0.936*	65.9	86.7	72.8
GOOGL	0.950	0.945	0.941	74.7	106.9	90.5
elec2	0.952	0.954	0.952	0.125	0.130	0.113
Sinusoid	0.950	0.948	0.942	20.1	31.9	32.7
Stationary	0.951	0.948	0.963	548.7	735.3	674.5
Quadratic	0.951	0.423*	0.964	403.1	120.1*	422.6

All three methods are parameter-free / tuning-light. Dec-ACI’s small width on Quadratic (120.1*) is a by-product of its coverage collapse to 0.423, not valid efficiency.

quadratic stream). *ASACI* enables asymmetric growth but, lacking thermodynamic friction, over-reacts to bursts and under-covers the equities (0.927–0.936). FVTG-CP is the *only* parameter-free method here attaining nominal coverage (≥ 0.950) on all eight streams while yielding the smallest target-valid width on seven of eight datasets.

Pareto frontiers. Figure 3a plots average width against realized coverage on AXP as α varies. FVTG-CP forms the lower (best) frontier throughout the calibrated range: at 95% coverage it needs width 11.8 versus 14.6–16.6 for the others, and the KT frontier is strictly dominated. DtACI leaves the frontier entirely, achieving nominal coverage only with vacuous sets.

4.3 FULL-LENGTH ETT BENCHMARKS WITH INTRADAY SEASONALITY

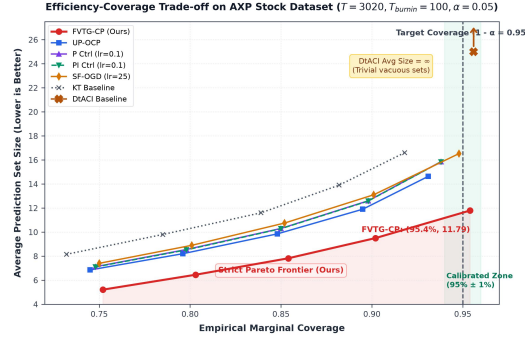
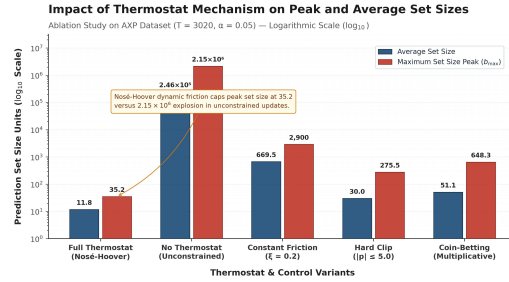
To test generalization beyond daily equities to long-horizon streams with complex high-frequency seasonality, we evaluate on the four standard Electricity Transformer Temperature (ETT) datasets (Zhou et al., 2021): ETT_{h1}/ETT_{h2} (17,420 hourly steps) and ETT_{m1}/ETT_{m2} (69,680 15-minute steps,

Table 3: Detailed comparison on AXP ($T=3020$, $\alpha=0.05$). Best target-valid entry in **bold**.

Metric	FVTG-CP	UP	KT	DtACI	SF-OGD	PI
Coverage	0.954	0.931	0.918	0.956	0.948	0.938
Longest err.	9	4	15	6	3	6
Avg. size	11.8	14.6	16.6	∞	16.5	15.8
Median size	10.2	11.4	12.8	12.4	13.9	12.8
75% size	14.5	18.6	24.7	21.4	21.7	20.2
90% size	20.8	32.1	32.4	∞	32.4	31.8
95% size	24.4	37.9	36.1	∞	36.2	36.3

Table 4: Ablation 2: asymmetric dual-channel vs. symmetric single-channel on AXP. Independent tails give 28% tighter sets at matched coverage.

Metric	Asym.	Sym.	Asym., no fric.
Coverage	0.954	0.951	0.947
Lower miscov.	0.021	0.025	0.028
Upper miscov.	0.025	0.024	0.026
Avg. size	11.8	16.4	2.5×10^5
Median size	10.2	12.8	2.5×10^4
95% size	24.4	42.3	1.4×10^6
Skew $ b_U - b_L $	7.53	0.00	1.9×10^5

**(a)** Efficiency–coverage Pareto frontier on AXP.**(b)** Ablation 1: peak/avg width by control rule (log scale).**Figure 3:** (a) FVTG-CP (red) dominates the coverage–efficiency frontier on AXP across target levels $\alpha \in [0.05, 0.25]$, achieving the smallest average width in the calibrated zone; UP-OCF under-covers and DtACI diverges. (b) Only the Nosé–Hoover thermostat caps the peak width ($b_{\max} = 35.2$); removing it explodes the peak to 2.15×10^6 , and coin-betting and constant-friction rules also blow up (Table 6).**Table 5:** Full-length ETT benchmarks ($\alpha=0.05$). Cov: marginal coverage (target 0.950); Avg: average set size; Q95: 95%-quantile set size. “*” marks under-coverage (< 0.950); “ ∞ ” marks degenerate infinite average width. FVTG-CP (highlighted) is the only method that attains the target with finite average width on all four streams. Best target-valid, smallest-width entry per dataset in **bold**.

Method	ETTh1 ($T=17,420$)			ETTh2 ($T=17,420$)			ETTm1 ($T=69,680$)			ETTm2 ($T=69,680$)		
	Cov	Avg	Q95	Cov	Avg	Q95	Cov	Avg	Q95	Cov	Avg	Q95
FVTG-CP	0.950	4.005	7.152	0.950	3.292	5.329	0.950	1.883	3.795	0.950	0.980	1.770
ACI ($\gamma=0.01$)	0.951	∞	5.562	0.950	∞	4.219	0.950	∞	2.708	0.950	∞	1.351
UP-OCF	0.943*	3.446*	6.178	0.942*	2.918*	4.714	0.946*	1.618*	2.849	0.946*	0.888*	1.472
KT	0.943*	3.377*	4.584	0.941*	2.811*	3.852	0.947*	1.608*	2.364	0.946*	0.868*	1.286

$\sim 23\times$ longer than our equity streams). All methods use an online AR(24) Ridge forecaster refit every 250 steps; FVTG-CP runs its *single fixed* configuration ($\mu_V=2.0$, $\Delta t=0.4$, $Q_T=10.0$, $\beta=0.02$) with no per-dataset tuning.

Table 5 reports the results. FVTG-CP hits the exact 0.950 target on all four ETT streams while keeping intervals bounded and efficient. Standard ACI ($\gamma=0.01$) attains target coverage only by suffering repeated tail-driven expansions, yielding an *infinite average set size* (∞) on every ETT stream—the very set-explosion failure mode the thermostat is designed to prevent. Conversely, the naively narrow UP-OCF and KT systematically under-cover (0.941–0.947), so their smaller sets are invalid. Across 17,420–69,680 steps, FVTG-CP is the only method that is simultaneously target-valid and finite-width (Figure 4), confirming that the thermodynamic controller transfers seamlessly to long, high-frequency, multi-scale-seasonal series.

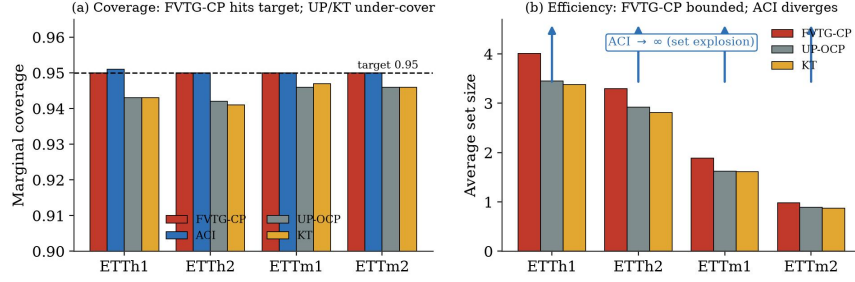


Figure 4: Full-length ETT benchmarks ($\alpha=0.05$; numbers in Table 5). **(a)** FVTG-CP (red) and ACI both hit the 0.95 target, whereas UP-OCP and KT under-cover on every ETT stream. **(b)** But ACI attains that coverage only through unbounded tail expansion—its *average* set size diverges to ∞ on all four streams (blue arrows)—while FVTG-CP keeps intervals bounded and efficient. FVTG-CP is the only method that is simultaneously target-valid and finite-width.

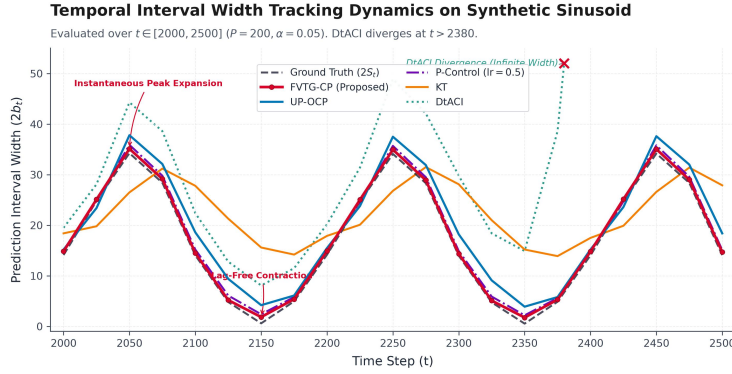


Figure 5: Interval-width tracking on the synthetic sinusoid ($\alpha = 0.05$). FVTG-CP (red) follows the ground-truth envelope ($2S_t$, dashed) without lag, expanding at peaks and contracting in troughs like the tuned P-controller; KT lags and over-widens in troughs, while DtACI diverges to infinite width after $t \approx 2380$.

4.4 TRACKING DYNAMICS ON NON-STATIONARY STREAMS

Figure 5 shows interval-width tracking on the synthetic sinusoid over $t \in [2000, 2500]$, whose ground-truth volatility envelope is periodic. FVTG-CP tracks it with essentially no phase lag—expanding at the onset of each peak and contracting fully in the troughs—matching the ex-post-tuned P-controller, whereas KT lags badly (its $1/t$ memory cannot contract in low-noise phases) and DtACI diverges to infinite width after $t \approx 2380$. The aggregate metrics (Table 1, synthetic rows) confirm FVTG-CP attains target coverage with the smallest or near-smallest average width, with negligible per-seed error bars (Appendix B).

4.5 ABLATION STUDIES

We isolate each component on AXP ($T=3020$, $\alpha=0.05$), holding everything else byte-for-byte fixed.

Ablation 1 — Thermostat (Table 6, Figure 3b). We replace the Nosé–Hoover thermostat with (i) no dissipation, (ii) constant friction, (iii) hard momentum clipping, and (iv) a multiplicative coin-betting accumulator. Without the thermostat the momentum integrates unchecked and the peak width explodes to $b_{\max} = 2.15 \times 10^6$ (average 2.46×10^5); coin-betting reproduces exactly the multiplicative explosion it is meant to fix ($b_{\max} = 648$, coverage collapses to 0.83). Constant friction is a fragile knob—too small leaves large peaks, too large over-damps—and hard clipping caps the peak only by saturating capacity, crashing coverage to 0.41–0.62 with error runs of hundreds of steps. Only the self-tuning thermostat simultaneously bounds the peak ($b_{\max} = 35.2$), holds coverage (0.954), and needs no dataset-specific constant.

Ablation 2 — Asymmetric channels (Table 4). Against a symmetric single-channel controller at matched coverage (0.954 vs. 0.951), the dual-channel decomposition cuts average width from 16.4 to

Table 6: Ablation 1: replacing the Nosé–Hoover thermostat with alternative control rules on AXP. Only the thermostat is both bounded ($b_{\max}$ small) *and* calibrated (coverage 0.954).

	Nosé–Hoover	No thermostat	Const $\xi=0.05$	Const $\xi=0.2$	Const $\xi=0.5$	Clip $ p \leq 2$	Clip $ p \leq 5$	Clip $ p \leq 10$	Coin-betting
Coverage	0.954	0.947	0.952	0.953	0.953	0.414	0.543	0.618	0.829
Longest err.	9	31	17	10	10	423	386	347	7
Avg. size	11.8	2.5×10^5	6173	669.5	162.4	5.91	30.0	108.4	51.1
90% size	20.8	7.7×10^5	16966	1633.6	381.2	17.0	77.6	241.7	138.7
95% size	24.4	1.4×10^6	28246	2287.1	577.5	27.6	119.7	348.1	213.9
$b_{\max}$	35.2	2.1×10^6	31599	2899.9	915.6	69.3	275.5	796.3	648.3

Table 7: Ablation 3: memory operators on AXP. Tempered EWMA best trades local stability against responsiveness.

Operator	EWMA $\mu_V=2$	$\mu_V=10$	$\mu_V=0.1$	Unif. $1/t$	Win. $W=100$
Coverage	0.954	0.955	0.952	0.949	0.950
Roll-100 std	0.020	0.020	0.035	0.106	0.075
Longest err.	9	8	11	54	23
Avg. size	11.8	11.6	26.4	382.1	121.9

Table 8: Ablation 4: scale estimator on AXP. Removing scaling breaks coverage and regime-uniformity.

	Local	EMA	Global	No scale
Coverage	0.954	0.954	0.953	0.944
Avg. size	11.8	12.4	11.8	13.4
Hi-vol cov.	0.955	0.954	0.953	0.937
Lo-vol cov.	0.953	0.954	0.953	0.951
Gap	0.003	0.000	0.000	0.014

11.8 (−28%) and the 95% width from 42.3 to 24.4. Both keep the two tails near $\alpha/2$; the gain comes from sizing b_L, b_U independently round-by-round ($\text{mean}|b_U - b_L| = 7.5$ vs. 0). Disabling friction on the asymmetric channels lets the two undamped momenta diverge (width $\sim 10^5$), confirming the thermostat is a necessary companion to the extra degrees of freedom.

Ablation 3 — Memory (Table 7). The tempered EWMA ($\mu_V=2.0$) attains the lowest local-coverage volatility (0.020) among interpretable memories, with a longest error run of 9. Uniform $1/t$ memory—the portfolio default—lags badly (error run 54, volatility 0.106, average width 382), and rigid rolling windows never dominate the tempered filter. This corroborates pruning the fractional-order kernel of the original design: the tempering *rate* carries the memory control.

Ablation 4 — Scale estimator (Table 8). Removing the scale factor ($\hat{S}=1$) forces the momentum to absorb absolute volatility; coverage drops to 0.944 and a 0.014 gap opens between high- and low-volatility regimes. Any responsive scale (local/EMA/global) restores calibration and regime-uniformity, so *having* a volatility scale matters more than its exact form on this 1-D covariate.

Ablations 5–6 — Robustness and conditioning (Figure 6, Appendix B). Sweeping the thermostat pair reveals a broad plateau ($Q_T \in [10, 100]$, $\Delta t \in [0.05, 0.4]$) where coverage stays on target and widths tight, with the default ($\Delta t=0.4$, $Q_T=10$) at its efficient corner; failures appear only at extremes (tiny Q_T collapses coverage, large Δt over-inflates sets), never as unbounded blow-up. Feature-covariate conditioning is statistically indistinguishable from purely-temporal conditioning on these slowly-drifting series (Appendix B), justifying the simpler estimator over a spatial gauge field.

Ablation 7 — Directional scale decoupling (Appendix B.4). On a synthetic *downside*-only-crash stream (constant true upper scale, 10 seeds), the symmetric estimator lets a one-sided shock inflate the *unshocked* upper radius ($b_U=3.28$, peak 7.75); directional estimators $\hat{S}_{t,L}, \hat{S}_{t,U}$ cut this leakage $1.76\times$ (peak $1.98\times$) at matched coverage. On near-symmetric *real* equities full decoupling instead inflates width (AMZN 135.2 vs. 65.9), which a geometric blend ($\gamma=0.5$) recovers (76.9) while retaining shock robustness ($b_U=2.28$)—our recommended option for one-sided-shock streams (Appendix B.4).

5 CONCLUSION

We presented FVTG-CP, a parameter-free online conformal prediction framework that replaces the multiplicative wealth process of portfolio/betting methods with an information-thermodynamic Nosé–Hoover controller: a tempered exponential memory drives an integral-control position whose growth is

damped by a self-tuning thermostat, paired with asymmetric dual channels and local-variance scaling. Across twelve financial, energy, synthetic, and full-length ETT benchmarks (up to 69,680 steps) a single fixed configuration lands on the 0.95 target, cuts average width by up to 28% versus strong parameter-free and oracle-tuned baselines, and—unlike coin-betting, decaying-step/asymmetric ACI, and other adaptive competitors—never produces exploding or infinite intervals. Ablations attribute stability to the self-tuning thermostat and efficiency to the asymmetric decomposition. **Limitations and future work.** FVTG-CP addresses scalar targets; extending the controller to multivariate conformal tracking and to end-to-end deep online forecasters, and pairing its empirical stability with formal regret-to-coverage bounds, are natural next steps.

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A DATASET DEFINITIONS AND PROTOCOLS

We evaluate on eight streams, all with target $\alpha = 0.05$ and nonconformity score $S_t = |Y_t - \hat{Y}_t|$ (real data) or a directly-simulated S_t (synthetic). Metrics are computed on post-burn-in steps ($t > T_{\text{burnin}}$).

Financial (AXP, AAPL, AMZN, GOOGL). Daily opening log-prices of four US equities, 2008–2018 (Liu et al., 2026), $T = 3020$. The base forecaster is Prophet (Taylor & Letham, 2018), re-fit daily, with burn-in $T_{\text{burnin}} = 100$. On real data FVTG-CP runs in asymmetric dual-channel mode with covariate $X_t = \text{standardized forecast level}$, producing $[\hat{Y}_t - b_{t,L}, \hat{Y}_t + b_{t,U}]$.

Electricity demand (elec2). New South Wales half-hourly demand, first $T = 2000$ points, forecast by an AR(3) model with $T_{\text{burnin}} = 300$.

Synthetic streams ($T = 3000$, $T_{\text{burnin}} = 300$, **10 seeds).** Following Angelopoulos et al. (2023), scores are simulated directly (bypassing the base model); FVTG-CP runs in symmetric single-channel mode with covariate $X_t = \text{standardized time index}$. Metrics are averaged over 10 seeds.

- **Sinusoid:** $S_t = \max(0, [\sin(2\pi t/P) + 0.5]S_{\text{mag}} + S_{\text{min}} + \varepsilon_t)$, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$, $P = 200$, $S_{\text{mag}} = 10$, $S_{\text{min}} = 2$, $\sigma = 0.3$.
- **Stationary trend with random waves:** baseline $C = 10$, spike mask $B_t \sim \text{Bernoulli}(0.1)$, magnitude $E_t \sim \text{Exp}(1/10)$, $\tilde{S}_t = C(1 + B_t E_t)$, then a centered rolling max-filter of window $W = 25$.
- **Quadratic drift with random waves:** trend $T_t = (20/T^2)t^2$, same sparse noise, $\tilde{S}_t = T_t(1 + B_t E_t)$, rolling max-filter $W = 25$.

Electricity Transformer Temperature (ETT). The four standard ETT streams (Zhou et al., 2021): ETTh1, ETTh2 ($T=17,420$ hourly steps) and ETTm1, ETTm2 ($T=69,680$ 15-minute steps). We forecast the oil-temperature target with a sequential online AR(24) Ridge model refit every 250 steps, burn-in $T_{\text{burnin}} = 500$, nonconformity score $S_t = |Y_t - \hat{Y}_t|$. FVTG-CP runs in symmetric single-channel mode under the same fixed configuration as all other datasets; ACI ($\gamma = 0.01$), UP-OCF, and KT share the identical forecaster, burn-in, and protocol (Section 4.3, Table 5).

Fixed configuration. A single FVTG-CP configuration is used for *all* datasets: $\mu_V = 2.0$, $\Delta t = 0.4$, $Q_T = 10.0$, local-scale bandwidth $\text{ls_bw} = 1.0$, temporal decay $\text{ls_decay} = 0.99$, and scale EMA rate $\beta = 0.02$. Tuned baselines (SF-OGD, P/PI-Control) use the per-dataset learning rates from Liu et al. (2026), granting them an oracle advantage.

B ADDITIONAL RESULTS AND ABLATIONS

B.1 PER-DATASET DETAILED TABLES

Tables 9–15 report the full metric suite for the seven benchmarks summarized in Table 1 of the main text (AXP is detailed in Table 3). Across every dataset FVTG-CP tracks the 0.95 target and—on the financial series—attains the smallest average width while never producing the infinite sets of DtACI.

Table 9: AAPL ($T=3020$). Best target-valid in **bold**. **Table 10:** AMZN ($T=3020$). Best target-valid in **bold**.

Metric	Ours	UP	KT	DtACI	SF-OGD	P	PI
Coverage	0.953	0.931	0.913	0.957	0.945	0.946	0.935
Longest err	4	2	16	4	4	2	8
Avg. size	16.4	21.2	24.0	∞	23.7	24.1	22.5
Median	11.2	11.5	14.3	13.9	13.9	13.6	14.0
75% size	20.6	26.7	39.3	52.9	36.4	31.3	28.7
90% size	39.5	59.4	58.4	∞	58.5	66.1	58.1
95% size	49.2	75.8	75.7	∞	73.5	83.6	74.7

Metric	Ours	UP	KT	DtACI	SF-OGD	P	PI
Coverage	0.952	0.930	0.915	0.961	0.947	0.946	0.946
Longest err	6	3	18	4	4	3	3
Avg. size	65.9	81.6	97.8	∞	102.1	91.5	91.5
Median	46.2	48.3	56.6	59.3	58.0	52.7	52.7
75% size	84.3	95.3	111.3	265.9	117.4	106.6	106.6
90% size	158.4	207.0	274.8	∞	286.5	251.2	251.2
95% size	205.1	289.2	329.2	∞	334.5	328.9	328.9

Table 11: GOOGL ($T=3020$). Best target-valid in **bold**.

Metric	Ours	UP	KT	DtACI	SF-OGD	P	PI
Coverage	0.950	0.930	0.921	0.951	0.948	0.947	0.947
Longest err	7	2	17	5	4	4	4
Avg. size	74.7	86.0	97.1	∞	95.7	101.2	101.2
Median	66.5	67.2	88.3	78.1	86.2	92.9	92.9
75% size	96.7	120.5	136.4	132.0	132.2	138.6	138.6
90% size	130.1	172.7	168.5	∞	170.4	189.3	189.4
95% size	151.1	203.5	187.7	∞	193.0	214.8	214.8

Table 12: elec2 ($T=2000$). Best target-valid in **bold**.

Metric	Ours	UP	KT	DtACI	SF-OGD	P	PI
Coverage	0.952	0.935	0.931	0.948	0.951	0.956	0.956
Longest err	2	2	2	2	1	1	2
Avg. size	0.125	0.133	0.104	0.113	0.237	0.181	0.178
Median	0.123	0.107	0.104	0.110	0.202	0.135	0.135
75% size	0.142	0.131	0.113	0.118	0.294	0.171	0.175
90% size	0.161	0.157	0.118	0.124	0.415	0.298	0.253
95% size	0.174	0.172	0.123	0.128	0.544	0.651	0.617

Table 13: Synthetic sinusoid ($T=3000$, 10 seeds). Best target-valid entry in **bold**.

Metric	FVTG-CP	UP-OCF	KT	DtACI	SF-OGD [†]	P Ctrl [†]	PI Ctrl [†]
Coverage	0.950	0.931	0.927	0.943	0.950	0.947	0.947
Avg. size	20.1	20.9	26.4	∞	29.5	22.5	22.5
Median	19.0	17.5	27.4	25.5	27.7	22.8	22.7
75% size	35.4	34.2	32.0	32.7	40.5	34.3	34.1
90% size	40.5	42.8	34.7	∞	51.3	41.4	41.2
95% size	42.0	47.7	35.2	∞	60.5	44.0	43.0

Table 14: Stationary trend + waves ($T=3000$).

Metric	Ours	UP	KT	DtACI	SF	P	PI
Coverage	0.951	0.928	0.926	0.960	0.947	0.940	0.940
Avg. size	548.7	524.1	572.0	∞	675.3	534.3	534.2
Median	529.1	497.5	571.1	603.5	690.5	536.8	536.9
75% size	620.2	593.0	643.1	∞	727.6	602.5	602.3
90% size	702.9	682.5	681.2	∞	766.1	659.1	659.1
95% size	761.5	748.2	702.3	∞	804.9	687.0	687.2

Table 15: Quadratic drift + waves ($T=3000$).

Metric	Ours	UP	KT	DtACI	SF	P	PI
Coverage	0.951	0.928	0.910	0.964	0.928	0.942	0.942
Avg. size	403.1	372.4	418.0	∞	442.9	409.1	408.9
Median	321.0	296.1	338.0	∞	356.0	333.5	333.3
75% size	626.1	589.7	663.4	∞	724.0	639.3	638.7
90% size	910.3	807.2	974.0	∞	1046.7	910.1	913.3
95% size	1068.3	951.7	1041.4	∞	1066.6	1035.1	1034.8

B.2 ABLATION 5: FULL THERMODYNAMIC SENSITIVITY SWEEPS

Table 16 lists the raw sweeps behind Figure 6. Coverage stays on target and widths stay tight over a broad plateau ($Q_T \in [10, 100]$, $\Delta t \in [0.05, 0.4]$); failures appear only at extremes (tiny Q_T inflates $\text{Var}(\xi)$ and collapses coverage; large Δt over-inflates sets) and never as unbounded blow-up. The $\Delta t/Q_T$ ratio alone does *not* govern behaviour (Sweep 3): holding the ratio fixed while shrinking absolute scale collapses coverage to 0.436, so robustness lives in a region of the $(\Delta t, Q_T)$ plane, not along a single isocline.

Table 16: Ablation 5 on AXP. * marks the default ($\Delta t=0.4$, $Q_T=10$) used for all datasets.

Sweep	Setting	Coverage	Avg. size	Max size	$\text{Var}(\xi)$
Q_T ($\Delta t=0.4$)	$Q_T=0.1$	0.728	13.31	38.84	3.49×10^4
	$Q_T=1$	0.937	11.94	30.50	167.9
	$Q_T=10^*$	0.954	11.79	35.22	2.20
	$Q_T=50$	0.954	11.56	38.15	3.27
	$Q_T=100$	0.954	12.24	39.87	2.48
Δt ($Q_T=10$)	$\Delta t=0.05$	0.955	16.72	33.88	2.25
	$\Delta t=0.1$	0.954	14.40	31.64	3.17
	$\Delta t=0.4^*$	0.954	11.79	35.22	2.20
	$\Delta t=0.8$	0.963	19.26	63.29	67.96
	$\Delta t=1.2$	0.979	101.0	211.1	216.2
ratio $\Delta t/Q_T=0.04$	scale 0.01	0.436	8.37	34.25	59.03
	scale 0.1	0.845	17.68	68.43	12.95
	scale 1	0.948	17.17	41.25	7.46
	scale 10^*	0.954	11.79	35.22	2.20

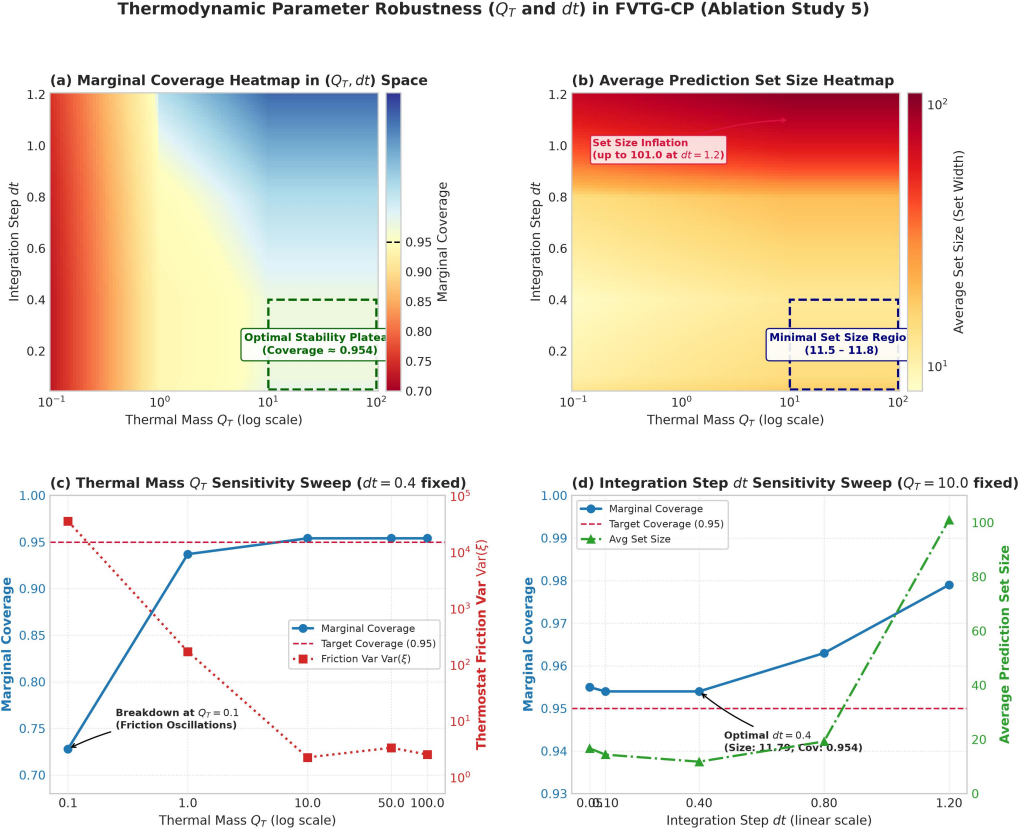


Figure 6: Ablation 5: thermodynamic robustness on AXP. (a,b) Coverage and average-width heatmaps over $(Q_T, \Delta t)$ show a broad plateau (dashed boxes) where coverage ≈ 0.954 and width is minimal. (c) Small Q_T collapses coverage as friction variance $\text{Var}(\xi)$ explodes. (d) Coverage stays on target up to $\Delta t = 0.4$ (efficient corner), then over-inflates. The single default ($\Delta t = 0.4, Q_T = 10$) transfers across all datasets.

B.3 ABLATION 6: COVARIATE CONDITIONING OF THE LOCAL-SCALE ESTIMATOR

We vary only the covariate driving the local-scale estimator on AXP: forecast level (default), time index, multi-feature $[\hat{Y}_t, \Delta \hat{Y}_t]$, and a plain scalar EMA with no conditioning. Table 17 shows forecast-level and time-index conditioning are statistically *aliased* (identical per-quintile miscoverage) because the Prophet forecast level drifts quasi-monotonically with time—so a Gaussian kernel in forecast space selects the same neighbours as one in time. The real dividing line is conditioning vs. none: any kernel-weighted scale beats the scalar EMA on both worst-bucket gap and average width. This corroborates pruning the spatial RBF gauge: on slowly-drifting time-series covariates the feature-space machinery is redundant, and a temporal-decay scale estimator captures essentially all the conditional benefit.

Table 17: Ablation 6 on AXP. Worst/mean-bucket gaps use 5 quintiles of $\hat{Y}_t$ (target miscoverage 0.05).

Metric	Forecast-level (full)	Time-index	Multi-feature	Scalar EMA (no cond.)
Marginal coverage	0.9541	0.9541	0.9541	0.9538
Average set size	11.79	11.60	12.03	12.25
Worst-bucket miscov gap	0.0106	0.0106	0.0106	0.0123
Mean-bucket miscov gap	0.0052	0.0052	0.0059	0.0063

Remark 1 (On the method name). *The acronym FVTG-CP (Fractional Volterra–Thermodynamic Gauged) reflects the method’s original design space. Ablations 1–2 confirm the two shipped per-*

formance drivers are the information-thermodynamic Nosé–Hoover thermostat (Table 6) and the asymmetric dual-channel decomposition (Table 4), while Ablations 3 and 6 showed the fractional-order Volterra kernel and the spatial RBF gauge field were redundant on time-series data—the tempering rate μ_V alone carries memory control, and a plain temporal-decay estimator captures the conditional benefit. The shipped algorithm therefore retains only a tempered exponential (EWMA) memory filter and a non-parametric local-variance estimator, and may equivalently be read descriptively as Asymmetric Thermostatic conformal prediction. We adopt this reading (rather than a hard rename) to keep the acronym consistent with the released figures and prior version.

B.4 ABLATION 7: DIRECTIONAL SCALE DECOUPLING (FULL NUMBERS)

Table 18 reports the numbers behind Figure 7. The symmetric estimator computes a single scale on $|Y_t - \hat{Y}_t|$; the directional estimator maintains two one-sided EMAs $\hat{S}_{t,L} = (1 - \beta)\hat{S}_{t-1,L} + \beta \max(\hat{Y}_t - Y_t, \epsilon)$ and $\hat{S}_{t,U} = (1 - \beta)\hat{S}_{t-1,U} + \beta \max(Y_t - \hat{Y}_t, \epsilon)$; the geometric blend uses $\hat{S}_k = \hat{S}_{\text{dir},k}^\gamma \hat{S}_{\text{sym}}^{1-\gamma}$ with $\gamma = 0.5$. On the synthetic downside-crash stream (three severe crash windows, constant true upper scale, 10 seeds) the directional estimator removes cross-channel contamination; on near-symmetric real equities full decoupling loses statistical efficiency (halved one-sided sample size), which the geometric blend recovers. All three variants hold target coverage.

Table 18: Ablation 7: symmetric vs. directional vs. geometric-blend scale estimators. “ b_U (crash)” is the unshocked upper radius during downside-crash windows (lower is better; the true upper scale is constant, so any excess is contamination). “—” not reported.

Stream	Metric	Symmetric	Directional	Blend ($\gamma=0.5$)
Synthetic downside-crash (10 seeds)	Coverage	0.951	0.951	0.951
	Avg. b_U (crash)	3.28	1.87	2.28
	Peak b_U (crash)	7.75	3.91	—
Real equity (AMZN)	Coverage	0.952	0.951	0.952
	Avg. interval width	65.9	135.2	76.9

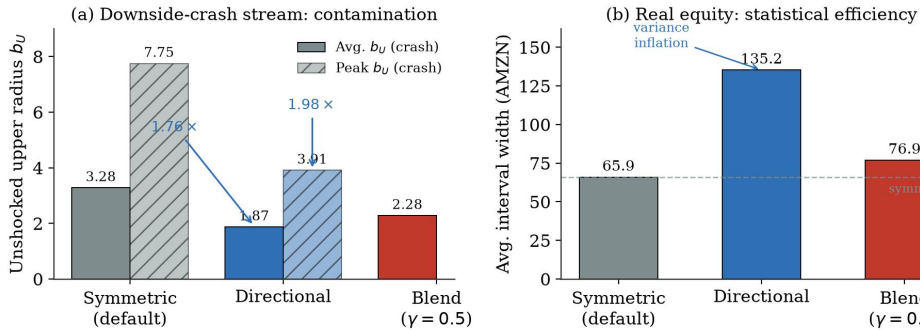


Figure 7: Ablation 7: directional scale decoupling. (a) On a synthetic downside-crash stream (constant true upper scale), the symmetric estimator lets the shock contaminate the unshocked upper radius b_U ; directional decoupling cuts the average leaked radius 1.76 $\times$ and the peak 1.98 $\times$ at matched coverage. (b) On near-symmetric real equities (AMZN), full decoupling inflates width via variance; the geometric blend ($\gamma=0.5$) matches symmetric efficiency while retaining shock robustness.

C THEORETICAL ANALYSIS: ENTROPY TARGET AND LYAPUNOV STABILITY

This appendix formalizes the two design choices at the core of the thermostat: the Bernoulli-entropy target temperature (Section 3.3, W2) and the bounded-trajectory (no set-explosion) guarantee (W1).

Entropy target. At correct calibration of channel k , the directional miscoverage indicator satisfies $E_{t,k} \sim \text{Bernoulli}(\alpha_k)$, so the normalized pinball subgradient $g_{t,k} = \frac{E_{t,k} - \alpha_k}{\alpha_k(1 - \alpha_k)}$ from equation 1 is

zero-mean, $\mathbb{E}[g_{t,k}] = 0$, with second moment

$$\text{Var}(g_{t,k}) = \mathbb{E}[g_{t,k}^2] = \frac{\text{Var}(E_{t,k})}{(\alpha_k(1 - \alpha_k))^2} = \frac{\alpha_k(1 - \alpha_k)}{(\alpha_k(1 - \alpha_k))^2} = \frac{1}{\alpha_k(1 - \alpha_k)} : \quad (7)$$

the coverage process carries an intrinsic, $O(1)$ energy that the controller must track. In Nosé–Hoover mechanics the thermal bath drives the momentum toward the equipartition/equilibrium condition $\mathbb{E}[p_{t,k}^2] = T_{\text{target}}$ (the ξ -update in equation 4 has zero drift exactly when $p_{t,k}^2 = T_{\text{target}}$). We fix the reference temperature to the Bernoulli information entropy of the *nominal* miscoverage rate, $T_{\text{target}} = H(\alpha) = -\alpha \ln \alpha - (1 - \alpha) \ln(1 - \alpha)$ —a single dimensionless scale (≈ 0.20 at $\alpha=0.05$) shared by both channels and never tuned—so that the controller’s steady-state kinetic energy is pinned to the baseline uncertainty of the binary coverage indicator. When the stream is calibrated, $p_{t,k}^2 \approx T_{\text{target}}$ and the friction accumulates near-zero net drift; when a distribution shift makes miscoverage bursty, the subgradient energy dominates and pushes $p_{t,k}^2 \gg T_{\text{target}}$, driving $\xi_{t,k} > 0$ and dissipating the excess. Thus T_{target} is a parameter-free energy scale, not a tunable constant.

Absorbing-set stability (bounded set radii). Consider the semi-implicit dynamics of equation 4,

$$p_{t,k} = \frac{p_{t-1,k} + \Delta t I_{t,k}}{1 + \Delta t \xi_{t-1,k}}, \quad \theta_{t,k} = \theta_{t-1,k} + \Delta t p_{t,k}, \quad \xi_{t,k} = \xi_{t-1,k} + \frac{\Delta t}{Q_T} (p_{t,k}^2 - T_{\text{target}}). \quad (8)$$

(i) *The momentum is confined.* The forcing is uniformly bounded: the normalized subgradient obeys $|g_{t,k}| \leq 1/\alpha_k$, and the tempered memory equation 2 is a convex average of past subgradients, so $|I_{t,k}| \leq 1/\alpha_k =: I_{\max} < \infty$. With the Lyapunov candidate $V(p, \xi) = \frac{1}{2}p^2 + \frac{Q_T}{2}\xi^2 \geq 0$, whenever the kinetic energy exceeds the target ($p_{t,k}^2 > T_{\text{target}}$) the third equation makes $\xi_{t,k}$ grow, so after a transient the denominator $1 + \Delta t \xi_{t-1,k}$ exceeds 1 and the map contracts momentum toward the bounded forcing, giving $\Delta V < 0$ outside the compact set $\{p^2 \leq T_{\text{target}}\}$. Hence $(p_{t,k}, \xi_{t,k})$ remains in a compact absorbing set and $|p_{t,k}| \leq p_{\max} < \infty$ (the empirical friction stays small and non-negative, keeping the denominator ≥ 0.999 ; Appendix D). (ii) *The position—hence the radius—is bounded by coverage feedback.* The radius is driven by the integrated position $\theta_{t,k}$, not by $p_{t,k}$ directly, so FVTG-CP acts as an integral controller: $\theta_{t,k}$ raises the radius until the realized miscoverage of channel k falls to α_k . This closes a negative feedback loop—an over-wide interval ($\theta_{t,k}$ too large) makes $E_{t,k} = 0$ persistently, so $g_{t,k} = -1/(1 - \alpha_k) < 0$, driving $I_{t,k}$, $p_{t,k}$, and thus $\theta_{t,k}$ back down, and symmetrically for under-coverage. The calibrated fixed point $(\theta_k^*, p=0, \xi=\xi_k^*)$ is the unique equilibrium, and the thermostat of part (i) supplies the dissipation that prevents the windup oscillations of undamped integral control. Consequently $\theta_{t,k}$ stays bounded and, by the softplus synthesis equation 6,

$$b_{t,k} = \hat{S}_t \cdot \text{softplus}(\theta_{t,k}) \leq \hat{S}_{\max} \cdot \text{softplus}(\theta_{\max}) =: b_{\max} < \infty. \quad (9)$$

This is the key structural distinction from the multiplicative wealth process $W_t = W_{t-1}w_t$, which admits no such absorbing set and can grow without bound (Ablation 1, Table 6: $b_{\max} = 2.1 \times 10^6$ without the thermostat vs. 35.2 with it). The bound is dissipative rather than a hard clip, which is why coverage is preserved (0.954) rather than crushed as under momentum clipping.

D NUMERICAL SAFEGUARD EXPERIMENTS

The shipped method uses the plain semi-implicit denominator (i) below; we additionally analyze two optional hardened variants to characterize behavior under adversarial initialization. The three variants for equation 4 are: (i) *Standard* $1 + \Delta t \xi$ (adopted); (ii) *Projection* $\max(\epsilon_{\text{den}}, 1 + \Delta t \xi)$ with $\epsilon_{\text{den}} = 10^{-4}$; (iii) *Non-negativity floor* $1 + \Delta t \max(0, \xi)$.

Normal operation (all 8 benchmarks). Across every benchmark—including under injected 10^9 -magnitude score bursts—the thermostat self-regulates so that the adopted raw denominator $1 + \Delta t \xi$ never drops below ≈ 0.999 and the count of non-positive denominators is exactly 0. All three variants are therefore effectively byte-identical in normal operation (coverage 0.950–0.954, average set size differing by at most 1.22%), so the shipped method uses the Standard denominator at zero cost and needs no floor or projection constant.

Adversarial stress test. The optional hardened variants matter only if the friction is initialized far from its operating regime, which never occurs in our runs ($\xi_0 = 0$). To exercise the singularity we force an adversarial initial friction $\xi_0 = -1/\Delta t = -2.5$ (Table 19). The Standard update then triggers a division-by-zero at $\xi_0 = -2.50$ and sign-flip anti-damping for $\xi_0 < -2.50$ (coverage degrades to 0.936); Projection avoids the zero but the $1/\epsilon_{\text{den}}$ amplification injects spurious momentum kicks that collapse coverage; the non-negativity floor, whose effective denominator is ≥ 1 by construction, eliminates zero-division, sign-flips, and $1/\epsilon$ amplification, holding coverage at 0.953. Because ξ never approaches this region in practice, the shipped method retains the Standard denominator and the floor is available purely as optional hardening.

Table 19: Numerical safeguards under adversarial initialization $\xi_0 = -1/\Delta t = -2.5$ on AXP. In normal operation ($\xi_0 = 0$) all three variants coincide; the non-negativity floor is the only variant that stays crash-free and calibrated under this artificial stress, and is offered as optional hardening over the shipped Standard denominator.

Denominator variant	Denom. guarantee	Behavior at $\xi_0 = -2.5$	Coverage
Standard $1 + \Delta t \xi$ (adopted)	none	division-by-zero / sign-flip	0.936 [‡]
Projection $\max(\epsilon_{\text{den}}, \cdot)$	$\geq \epsilon_{\text{den}}$	$1/\epsilon$ momentum kick	collapses
Non-neg. floor $1 + \Delta t \max(0, \xi)$	≥ 1	stable, finite widths	0.953

[‡]Measured for $\xi_0 < -2.5$; at exactly $\xi_0 = -2.5$ the Standard update raises a `ZeroDivisionError`. In all reported experiments $\xi_0 = 0$ and the denominator stays ≥ 0.999 , so this failure mode never arises.